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$$\boxed{\sigma(z) = \frac{1}{1+e^{-z}}}$$

CS m146 HW#2

1. $J_{hinge}(w,b) = \sum_{n=1}^{N} \max(0, 1 - y_n(w^T x_n + b)) + \frac{\lambda}{2}\|w\|_2^2$ $y_n \in \{+1, -1\}$, $\lambda > 0$

logistic loss $J_{log}(w,b) = \sum_{n=1}^{N} \log(1 + e^{-y_n(w^T x_n + b)}) + \frac{\lambda}{2}\|w\|_2^2$, $m_n = y_n(w^T x_n + b)$ = margin of nth

a.I. $\ell_{log}(w,b) = \log(1 + e^{-y(w^T x + b)})$ let $m = y(w^T x + b)$ = margin

$= \log(1 + e^{-m})$ $\frac{\partial}{\partial m} = \frac{-e^{-m}}{1+e^{-m}} \cdot yx$ $\boxed{\nabla_w \ell_{log}(w,b) = \frac{-e^{-y(w^T x + b)}}{1+e^{-y(w^T x + b)}} \cdot yx} = \frac{-e^{-m}}{1+e^{-m}} \cdot yx = \frac{-yx}{e^{m}+1}$

a.II. $\frac{\partial}{\partial b} \ell_{log}(w,b) = \frac{-e^{-m}}{1+e^{-m}} \cdot y = \frac{-e^{-m}}{1+e^{-m}} \cdot \frac{e^{m}}{e^{m}} = \boxed{\frac{-y}{1+e^{m}}} = \boxed{-\sigma(-m) \cdot y}$

b. $\boxed{\nabla_w J_{log} = \sum_{n=1}^{N} \frac{-y_n x_n}{1+e^{m_n}} + w\lambda}$ $\frac{\partial}{\partial w} \frac{\lambda}{2}\|w\|^2 = \frac{\lambda}{2} 2w = \lambda w$ $\boxed{\frac{\partial}{\partial b} J_{log}(w,b) = \sum_{n=1}^{N} \frac{-y_n}{1+e^{m_n}} = -\sum_{n=1}^{N} \sigma(-m_n) y_n}$

C. $\nabla_w \ell_n = \frac{-e^{-m_n}}{1+e^{-m_n}} y_n x_n$ mag $= \frac{e^{-m_n}}{1+e^{-m_n}} = \frac{1}{1+e^{m}}$ is between 0 & 1

if $m >> 0$ then mag $\to 0$, if $m \approx 0$ mag $\to \frac{1}{2}$ $m << 0$, mag $\to 1$ So Inversley Corelated

I A correctly classified example far away from desision boundary doesn't influence the weight

II. $m \approx 0$ - close to boundary so midpoint of 1/2 & unsure so pull harder on the boundary

III. $m << 0$ cofindently miss classified → so badly classified contributes the strongest mag is bounded at 1 no matter how wrong

d. $\ell_{hinge}(w,b) = \max(0, 1 - y(w^T x + b))$ max the margin ; logistic $\nabla \log = \frac{-y}{1+e^{m}}$

I. $m \geq 1 : 0$ $\Rightarrow$ $\nabla_w \text{hinge} \begin{cases} 0 & m \geq 1 \\ -yx & m < 1 \end{cases}$ Hinge loss contributes to zero gradient when $m \geq 1$

$m < 1 : 1 - m$

II. No it can not but the denominator can get very big

III. correctly classified with large positive margin => that is log loss. In hinge its exactly 0.

E. $m << 0$ so confidently missclasified & magnitude is bounded at 1 for log.

For hinge if $m < 0$ then $\nabla_{hinge} \ell = -yx$ so it is not dependent on m & no matter how missclassified you are hinge loss treats them the same, $m = .9 \to -1000$ doesn't matter. They both reach the same because $-yx$ = x is 1 & $-y = + -1$ so both approach 1.

F. $h(x) = \text{sign}(w^T x + b)$ replace hinge with log change boundary or training objective?

The Boundary is defined on where it switches sign : $w^T x + b = 0$ → is dependent on equasion not how w & b are learned

But they do produce different values of w & b because hinge only when $m < 1$ only listens to support vectors while log listens to every vector.

$k(x,z) = \phi(x)^T \phi(z)$

2. Gausian RBF kernel $= \exp(-\|x - z\|^2)$, $x, z \in \mathbb{R}^D$ a. $\|x - z\|^2 = (x - z)^T(x - z) = x^T x - 2x^T z + z^T z$

$k(x,z) = \exp(-(x^T x - 2x^T z + z^T z)) = \exp(-x^T x + 2x^T z - z^T z)$ ← only dot prods

b. $d_k(x,z) = \|\phi(x) - \phi(z)\|^2 = (\phi(x) - \phi(z))^T(\phi(x) - \phi(z)) = \phi(x)^T\phi(x) - 2\phi(x)^T\phi(z) + \phi(z)^T\phi(z)$

$= k(x,x) - 2k(x,z) + k(z,z)$ = kernel trick

II. $k(x,x) = \exp(-\|x - x\|^2) = \exp(0) = 1$ ← every point of $\phi(x)$ lives on unit sphere so $\|x\|^2 = 1$

III. $d_k(x,z) = 2 - 2k(x,z)$

Hw #2 Pt. 2

2. c. $d_k(x, z) = 2 - 2\,k(x, z)$ where $k(x, z) = \phi(x)^T \phi(z) = \exp(-\|x - z\|^2)$ $x, z \in \mathbb{R}^d$

I. $\|x - z\| \to 0$ then $k(x, z) = 1$ and $d_k(x, z) = 0$ points that are close in input are close in feature

II. $\|x - z\| \to \infty$ then $k(x, z) \Rightarrow 0$ $d_k(x, z) = 2$ no matter how far apart x & z they're feature space dist is capped at 2.

Since they all sit on unit sphere max dist is 2 but not mapped to a bounded region x, z ⊥

Classifier of this form d. $h(x) = \sum_{n=1}^{N} \alpha_n k(x_n, x)$ $h(x)$? when test point x is far from all training points $\{x_n\}_1^N$?

$\|x - x_n\| \gg 0$ for every n so $k(x_n, x) \Rightarrow 0$ therefore $h(x) = \sum_1^N \alpha_n (k\, x_n, x) \to 0$

model becomes uncertain and says I dont know for this classifier ✓

E. Everything we need was computed using $k(x, z)$ & never $\phi(x)$ directly

ex: feature space distances $k(x, x) + k(z, z) - 2k(x, z)$, Predictions $\sum \alpha_n k(x_n, x)$, and Training $k_{ij} = k(x_i, x_j)$

RBFS kernel feature space $\in \mathbb{R}^\infty$ taylor expansion of $e^{-\|x - z\|^2}$ but kernel is scalar $O(d)$ to calc.

3. $k(x, z) = \|x - z\|^2$ $x, z \in \mathbb{R}^d$ a. assume $k(x, z) = \phi(x)^T \phi(z)$

$d_k(x, z) = \|\phi(x) - \phi(z)\|_2^2$ I. $= (\phi(x) - \phi(z))^T(\phi(x) - \phi(z)) = \phi(x)^T\phi(x) - 2\phi(x)^T\phi(z) + \phi(z)^T\phi(z)$

$= k(x, x) + k(z, z) - 2k(x, z)$ II. $k(x, x) = 0$ ⇐ every point maps to zero vector?

III. $d_k(x, z) = -2\,k(x, z) = -2\|x - z\|^2$

b. d_k can not be a valid squared dist because its always neg & squared dist should be +,

Also kernel measures similarity so $k(x, x)$ should be pretty similar and not 0.

c. $K = \begin{bmatrix} k(x_1, x_1) & k(x_1, x_2) \\ k(x_2, x_1) & k(x_2, x_2) \end{bmatrix}$ I. compute K, $x_1, x_2 \in \mathbb{R}^d$ $\delta = \|x_1 - x_2\|^2 > 0$

then $k(x_1, x_1) = 0$ $k(x_1, x_2) = \|x_1 - x_2\|^2 = \delta = \begin{bmatrix} 0 & \delta \\ \delta & 0 \end{bmatrix}$

II. $\begin{bmatrix} -\lambda & \delta \\ \delta & -\lambda \end{bmatrix}$ $\lambda^2 - \delta^2 = 0$ $\lambda = \sqrt{\delta^2}$ $\lambda = \pm\delta$

III Is k PSD? meaning all eigenvals are ≥ 0 but this is not true $\lambda_2 = -\|x_1 - x_2\|^2 < 0$

d. $k(x, z) = \|x - z\|^2$ is not valid kernel because

1. $d_k(x, z) < 0$ This must be ≥ 0
2. $\lambda = -\delta < 0$ This also must be ≥ 0 All eigenvalues of gram matrix must be ≥ 0. & all gram matrixes must be PSd.
3. Squared euclidian distance shows that kernel should be large when points are close because it measures similarity!